Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **distributed backends** and optimizing **latency** and **reliability**; project experience in **C++** scheduling, with Linux and concurrent data processing skills.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Reduced dashboard latency from 1.2s to 350ms through **query optimization** and indexing on Azure SQL, architecting normalized multi-tenant storage and row-level isolation supporting 500+ users on an education platform.
- Tripled concurrent throughput by restructuring **fault-tolerant asynchronous** FastAPI workers, processing 1K+ AI jobs with explicit state transitions and failure isolation to prevent cascading task failures under concurrent production workloads.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Enabled **concurrent batch processing** of 1K+ files using Python multithreading, secure ingestion, REST APIs, and AWS RDS, supporting fault-tolerant execution across PDF, DOCX, and ZIP assessment uploads for instructors.
- Reduced feedback turnaround by 60% through **production backend** and CI/CD workflows serving 150+ instructors and 10K+ daily grading requests, maintaining stable API behavior during peak assessment processing loads.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Improved long-horizon planning by 18% in simulation using a Decision Transformer, building **data processing pipelines** for 100K+ trajectory logs to support offline reinforcement learning experiments and behavioral planning evaluation.
- Improved closed-loop planning stability by 15% through a **spatial-temporal encoder** combining HD-map geometry and historical ego-agent context, using multi-head attention and simulation evaluation to assess trajectory behavior across experiments.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Enabled reliable **cross-platform delivery** across Linux, Windows, and macOS through Jenkins-driven executable builds, automated VM installation, and 20+ unit/integration test suites, validating synchronization workflows with PyTest and Mockito during releases.
- Improved API response times by 30% using **object-oriented design**, indexing, caching, and payload filtering; led AWS Glue and S3 pipeline improvements that increased efficiency by 68% while reducing technical debt.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Delivered **sub-200ms booking APIs** using Flask and Firebase across 50K+ booking records, implementing pricing, coupon validation, payment webhooks, and real-time status synchronization for a cross-platform house-help service application backend.
- Improved booking confirmation speed by 40% through **real-time synchronization**, using Firebase transactions and locking to prevent double bookings while automating refund and payment-status workflows to reduce manual handling by 70%.

Skills

Languages & Systems: C++, Python, Java, SQL, Bash, Linux, TCP/IP, Operating Systems, Object-Oriented Design
Backend & Concurrency: Distributed Systems, System Design, Multithreading, Asynchronous Processing, Fault Isolation, REST APIs, FastAPI, Flask, Microservices
Data & Performance: Performance Optimization, Query Optimization, Indexing, Caching, PostgreSQL, MySQL, Azure SQL, ETL Pipelines, Data Validation, Data Modeling
Cloud & Delivery: AWS S3, AWS Glue, AWS RDS, Microsoft Azure, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD
Testing & Tools: Git, PyTest, Mockito, Unit Testing, Integration Testing, VM Automation, WebSockets, JSON, Agile

Projects

Event Stream Replay - Design Concept

- Proposed a **C++ event replay** architecture with ordered message processing, bounded buffering, and recovery checkpoints, outlining reproducible throughput and latency evaluation for financial data streams under varying simulated workloads.

Quantum-Resistant Blockchain | Paper

- Evaluated **throughput and latency** across 100-1,000 transactions in a FALCON-signature blockchain framework, sustaining 957.8 TPS at 224ms versus 968.3 TPS at 211ms for the ECDSA baseline, quantifying protocol trade-offs.

CPU-scheduling Simulator | Live

- Implemented **C++ scheduling algorithms** including FCFS, SJF, SRT, Round Robin, priority, and MLQ, validating preemptive and non-preemptive behavior with interactive visualization and waiting-time calculations across 1K+ simulated process combinations.